

While Loop

Syntax

```
In [1]: while condition==True:
        statement 1
        statement 2
        ...
        statement3
```

Cell In[1], line 2

statement 1

^

SyntaxError: invalid syntax

```
In [ ]: i=1
        while i<5:
            print(f"i=={i}")
            i+=1
```

```
In [ ]: s1='python'
        i=0
        while i<len(s1):
            print(f"Character at index {i} is {s1[i]}")
            i+=1
```

```
In [ ]: s1='python'
        i=0
        while i<len(s1):
            print(f"Character at index {i} is {s1[i]}")
            print("Machine Learning")
            i+=1
```

```
In [1]: n=10
        i=1
        while i<=n:
            print(i)
            i+=1
```

1
2
3
4
5
6
7
8
9
10

```
In [7]: n=10
i=1
sum1=0
while i<=n:
    print(i)
    sum1+=i
    i+=1
print("Sum of numbers is:",sum1)
```

```
1
2
3
4
5
6
7
8
9
10
Sum of numbers is: 55
```

```
In [6]: n=10
i=1
sum1=0
while i<=n:
    sum1+=i
    i+=1
    print("Sum of numbers is:",sum1)
```

```
Sum of numbers is: 1
Sum of numbers is: 3
Sum of numbers is: 6
Sum of numbers is: 10
Sum of numbers is: 15
Sum of numbers is: 21
Sum of numbers is: 28
Sum of numbers is: 36
Sum of numbers is: 45
Sum of numbers is: 55
```

```
In [8]: n=5
i=1
fact=1
while i<=n:
    fact*=i
    i+=1
print(f"Factorial of {n} is :",fact)
```

```
Factorial of 5 is : 120
```

```
In [13]: n=int(input("Enter the number="))
i=1
fact=1
if n>0:
    while i<=n:
        fact*=i
        i+=1
    print(f"Factorial of {n} is :",fact)
else:
    print("Factorial does not exist")
```

Enter the number=10
Factorial of 10 is : 3628800

```
In [14]: n=int(input("Enter the number="))
i=1
fact=1
if n>0:
    while i<=n:
        fact*=i
        i+=1
    print(f"Factorial of {n} is :",fact)
else:
    print("Factorial does not exist")
```

Enter the number=-5
Factorial does not exist

Loop Control Statements

```
In [ ]: 1)break
        2)continue
        3)pass
```

1)break

In []: It **is** used to terminate **all** operations of loop.

```
In [15]: for i in range(5):
          print(i)
          break
          print("Hello")
```

0

```
In [16]: i=5
          while i<=5:
              print(i)
              i+=1
```

5

```
In [17]: i=5
while i<=10:
    print(f"i=={i}")
    i+=1
    if i==7:
        print("Number found")
```

```
i==5
i==6
Number found
i==7
i==8
i==9
i==10
```

```
In [18]: i=5
while i<=10:
    print(f"i=={i}")
    i+=1
    if i==8:
        print("Number Found")
```

```
i==5
i==6
i==7
Number Found
i==8
i==9
i==10
```

```
In [19]: i=5
while i<=10:
    print(f"i=={i}")
    i+=1
    if i==7:
        break
    print("Number found")
    print("*"*30)
```

```
i==5
*****
i==6
```

```
In [20]: i=5
while i<=10:
    print(f"i=={i}")

    if i==7:
        print("Number found")
    i+=1
    print("*"*30)
```

```
i==5
*****

i==6
*****

i==7
Number found
*****

i==8
*****

i==9
*****

i==10
*****
```

```
In [21]: i=5
while i<=10:
    print(f"i=={i}")

    if i==7:
        break
        print("Number found")
    i+=1
    print("*"*30)
```

```
i==5
*****

i==6
*****

i==7
```

2)continue

In []: It **is** used to terminate current operations (iterations) only.

```
In [22]: i=11
while i<=20:
    print(f"i=={i}")

    if i==15:
        print("Hello")

    i+=1
    print("*"*30)
```

```
i==11
*****

i==12
*****

i==13
*****

i==14
*****

i==15
Hello
*****

i==16
*****

i==17
*****

i==18
*****

i==19
*****

i==20
*****
```

```
In [23]: i=11
while i<=20:
    print(f"i=={i}")
    i+=1
    if i==15:
        print("Hello")
        continue

    print("*"*30)
```

```
i==11
*****

i==12
*****

i==13
*****

i==14
Hello
i==15
*****

i==16
*****

i==17
*****

i==18
*****

i==19
*****

i==20
*****
```

```
In [24]: i=11
while i<=20:
    print(f"i=={i}")
    i+=1
    if i==15:
        print("Hello")
        continue
        print("Data Science")
        print("Machine learning")

    print("*"*30)
```

```
i==11
*****
i==12
*****
i==13
*****
i==14
Hello
i==15
*****
i==16
*****
i==17
*****
i==18
*****
i==19
*****
i==20
*****
```



```
In [25]: i=11
while i<=20:
    print(f"i=={i}")
    i+=1
    if i==15:
        print("Hello")
        print("Data Science")
        continue

    print("Machine learning")

print("*"*30)
```

```
i==11
*****
i==12
*****
i==13
*****
i==14
Hello
Data Science
i==15
*****
i==16
*****
i==17
*****
i==18
*****
i==19
*****
i==20
*****
```

pass

```
In [29]: i=11
while i<=20:
    print(f"i=={i}")
    i+=1
    if i==15:
        pass
```

```
File <tokenize>:7
    print("*"*30)
    ^
```

IndentationError: unindent does not match any outer indentation level

while else

```
In [30]: for i in range(5,10):
          print(i)
        else:
          print("We are completing all iterations")
```

```
5
6
7
8
9
We are completing all iterations
```

```
In [31]: i=5
        while i<=10:
            print(f"i=={i}")
            if i==8:
                break
            i+=1
        else:
            print("We are completing all iterations")
```

```
i==5
i==6
i==7
i==8
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

```
In [32]: n=5
        for i in range(5):
            for j in range(n):
                print("*",end=' ')
            print()
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

```
In [33]: n=5
i=0
while i<n:
    i+=1
    j=0
    while j<n:
        print("*",end=' ')
        j+=1
    print()
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

```
*
* *
* * *
* * * *
* * * * *
```

```
In [34]: n=5
i=0
while i<n:
    i+=1
    j=0
    while j<i:
        print("*",end=' ')
        j+=1
    print()
```

```
*
* *
* * *
* * * *
* * * * *
```

```
In [35]: n=5
i=0
while i<n:
    i+=1
    j=0
    while j<i:
        print("A",end=' ')
        j+=1
    print()
```

```
A
A A
A A A
A A A A
A A A A A
```

```
A
B C
D E F
```

```
G H I J  
K L M N O
```

```
In [36]: n=5  
i=0  
x=65  
while i<n:  
    i+=1  
    j=0  
    while j<i:  
        print(chr(x),end=' ' )  
        j+=1  
        x+=1  
    print()
```

```
A  
B C  
D E F  
G H I J  
K L M N O
```

```
In [ ]:
```